

# Melody-to-Piano: Transformer-Based Full Piano Accompaniment Generation from Simple Melody

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## Abstract

This project studies automatic piano accompaniment generation: given a monophonic melody, generate a full two-hand piano arrangement. The task is formulated as a conditional sequence-to-sequence problem over symbolic MIDI tokens. We evaluate whether a Transformer encoder-decoder can generate harmonically coherent and rhythmically structured accompaniment from melody input. Using the POP909 dataset, we compare a rule-based baseline, a melody-only Transformer, and a chord-conditioned Transformer. Evaluation is conducted using objective symbolic metrics, including note-level F1, onset F1, chord-tone ratio, and register balance statistics.

## 1 Problem Statement and Motivation

Given a simple melody line, the goal is to generate a full piano accompaniment that includes both left- and right-hand parts. Unlike unconditional music generation, this task requires structured prediction: the accompaniment must remain harmonically consistent with the melody while producing rhythmically meaningful polyphonic texture.

The problem is analogous to machine translation, where the input sequence (melody) must be mapped to a structured output sequence (piano accompaniment). The central research question is:

Can a Transformer-based sequence-to-sequence model generate harmonically coherent and rhythmically structured piano accompaniment from melody input?

We further evaluate whether explicit chord conditioning improves harmonic stability and structural consistency.

## 2 Related Work

Symbolic music generation has been studied using both recurrent neural networks and Transformer-

based architectures. LSTM-based models demonstrated early success in modeling temporal dependencies in music, while the Music Transformer introduced relative attention mechanisms to better capture long-term structure in symbolic sequences (Huang et al., 2018).

Most prior work focuses on unconditional generation or continuation tasks. In contrast, conditional accompaniment generation from melody remains less explored and can be framed as a structured sequence modeling problem. The POP909 dataset provides aligned melody and piano accompaniment tracks that are well suited for supervised learning (Wang et al., 2020).

## 3 Approach

### 3.1 Task Formulation

Let  $M$  denote a melody sequence and  $P$  the corresponding piano accompaniment. The task is modeled as:

$$P = f_{\theta}(M)$$

where  $f_{\theta}$  is a neural encoder-decoder model.

### 3.2 Representation

MIDI files are converted into event-based token sequences using:

- NOTE\_ON(pitch)
- NOTE\_OFF(pitch)
- TIME\_SHIFT(duration)
- VELOCITY(level)

Training is performed on fixed 4–8 bar segments to ensure stable optimization and manageable sequence length.

### 3.3 Models

We implement three systems:

#### Baseline: Rule-Based Generator

A deterministic accompaniment generator using chord labels to produce block chords or simple arpeggiated patterns.

#### Model A: Transformer (Melody → Piano)

A small Transformer encoder–decoder that maps melody tokens to accompaniment tokens.

#### Model B: Transformer + Chord Conditioning

Chord tokens are concatenated to melody tokens in the encoder input to evaluate the effect of explicit harmonic conditioning.

All neural models are implemented in PyTorch and trained with cross-entropy loss using teacher forcing.

## 4 Experimental Setup

### 4.1 Dataset

We use the POP909 dataset, which contains 909 pop songs with aligned melody, piano accompaniment, tempo, and chord annotations (Wang et al., 2020). The dataset will be split 80% train, 10% validation, and 10% test.

### 4.2 Training Details

- 4–8 bar training segments
- Adam optimizer
- Early stopping on validation loss
- Modest model size for stable convergence

### 4.3 Evaluation Metrics

We evaluate using objective symbolic metrics:

- Note-level Precision / Recall / F1
- Onset F1
- Chord-tone ratio (harmonic consistency)
- Register distribution statistics (left vs. right hand balance)

Qualitative MIDI renderings will be included for analysis.

### 4.4 Comparisons

We compare:

1. Rule-based baseline
2. Transformer (melody only)
3. Transformer + chord conditioning

This isolates the impact of attention mechanisms and harmonic conditioning.

## 5 Timeline and Work Breakdown

### Milestone (Mar 19):

- Data preprocessing complete
- Baseline implemented
- Melody-only Transformer trained
- Preliminary evaluation results

### Final Phase (Apr 9):

- Chord-conditioned model implemented
- Full evaluation completed
- Error analysis and qualitative examples

### Work Allocation:

- Mohammad: Model implementation, training, evaluation
- Sydney: Data preprocessing, baseline implementation, related work

## References

- Cheng-Zhi Anna Huang, Ashish Vaswani, Jakob Uszkoreit, Noam Shazeer, Ian Simon, Curtis Hawthorne, Andrew M. Dai, Matthew D. Hoffman, Monica Dinculescu, and Douglas Eck. 2018. [Music transformer](#). *Preprint*, arXiv:1809.04281.
- Ziyu Wang, Ke Chen, Junyan Jiang, Yiyi Zhang, Mao-ran Xu, Shuqi Dai, Guxian Bin, and Gus Xia. 2020. Pop909: A pop-song dataset for music arrangement generation. In *Proceedings of 21st International Conference on Music Information Retrieval, ISMIR*.